

# Usability Evaluation of the XJTLU Museum AI System

A simple task-based study with 12-15 XJTLU students or staff.



SYSTEM

XJTLU Museum AI system

An AI-supported visitor guide for exhibit information and museum navigation.

CONTEXT

Museum visitor guidance

Starting a tour, asking questions, navigating stops, and recovering from unclear answers.

PARTICIPANTS

12-15 campus visitors

Students or staff; one 15-minute task-based session.

METHOD

Task-based usability test

Four tasks, think-aloud comments, SUS, and a short debrief.

01 PROBLEM & MOTIVATION

First-time visitors may struggle to start, navigate, and understand the Museum AI system.

Gap: there is no public evidence on how visitors actually complete the Museum AI tasks.

02 QUESTIONS & HYPOTHESES

- RQ1: Can visitors complete the core tour tasks?
- RQ2: How usable and efficient is the system?
- RQ3: What barriers do visitors report?
- H1: At least 80% independent task success.
- H2: SUS meets the usability benchmark.

03 EXPERIMENT DESIGN

Each participant completes four core tasks with the Museum AI system.

1

Consent

Purpose + rights

2

Warm-up

System overview

3

Tasks

Four scenarios

4

SUS

10 ratings

5

Debrief

Barriers + ideas

04 MEASURES & SAMPLE

Task success

Independent / assisted / failed

Time on task

Seconds per task

Errors & help

Count by task

Usability

10-item SUS

Perceived ease

1-7 rating

Trust

Confidence in guidance

Sample: 12-15 XJTLU students or staff; one session per participant.

05 PRINCIPAL FINDINGS

Task success

Four-task profile

Efficiency

Time and errors

Usability

SUS, ease, and trust

Barriers

Observed and reported themes

Pending data: Insert real figures and anonymised evidence after data collection.

06 LIMITS & RECOMMENDATIONS

Limit

Small sample and controlled setting

Onboarding

Clarify how to start the tour.

Navigation

Improve route feedback.

Recovery

Add retry and correction paths.

07 REFERENCES

- XJTLU Museum, technology and AI system, 2026.
- ISO 2441-11: Usability definitions, 2018.
- S. Ameshi et al., "Guidelines for human-AI interaction," CHI, 2019.
- J. Brooke, SUS usability scale, 1996.

Acknowledgement: The evaluated system is the XJTLU Museum AI-supported visitor system. No ownership of the original system is claimed.

ADS401 Evaluation Methods and Statistics - GroupXX

Current draft: no participant data are presented.

A1 portrait: 59.5 × 84.1 cm